

An Analytical Study of Hospital Characteristics and Capacity in Maryland

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1. Introduction

1.1 Motivation and Research Questions

Access to healthcare facilities represents a fundamental dimension of public health infrastructure that directly impacts population health outcomes, emergency response capabilities, and healthcare equity. The geographic distribution of hospital resources, particularly licensed bed capacity, creates spatial patterns of healthcare accessibility that may systematically advantage some populations while disadvantaging others. These disparities often manifest along urban-rural divides, where rural populations frequently face longer travel times, fewer specialized services, and reduced hospital capacity per capita compared to their urban counterparts.

Maryland presents a particularly interesting case study due to its pronounced socioeconomic and geographic diversity. The state contains major urban centers (Baltimore City, the suburbs of Washington D.C.), extensive suburban areas, agricultural regions, and remote mountainous areas. This variation creates a natural laboratory for examining how hospital infrastructure aligns with population distribution and needs.

This project employs statistical methods to systematically analyze the distribution of hospital capacity across Maryland and identify factors associated with variations in licensed bed counts. We address these specific research questions that are derived from a healthcare equity concern standpoint:

1. Is there a relationship between hospital ownership and the number of specialized services offered?
2. Are there regional disparities in the availability of acute vs. special hospitals?
3. Is there any correlation between hospital type and ownership?
4. What is the average bed-to-hospital ratio by county, and how does it compare across regions?
5. Which MD counties are most vulnerable to healthcare capacity crises based on the combination of low facility density, low licensed capacity, and distance to nearest alternative facilities?

6. Using spatial clustering analysis, where do "healthcare deserts" exist in Maryland, and what is the average travel distance for residents in these areas to reach licensed hospital capacity?
7. What percentage of Maryland's population lives within a 15-minute drive of a hospital facility with more than 50 licensed beds, and how does this vary by county income levels?
8. Do federally designated Medically Underserved Areas (MUAs) in Maryland have systematically lower hospital capacity per capita, and what is the magnitude of this disparity?

The practical motivation for this analysis stems from its potential applications in public health planning, emergency preparedness, and resource allocation. Quantifying existing disparities provides evidence for targeted interventions, while understanding predictors of hospital size informs infrastructure planning decisions. From a methodological perspective, this project demonstrates how statistical techniques can transform administrative data into actionable insights for public healthcare infrastructure.

1.2 Data Background

The dataset used in this analysis is *“Acute, General, and Special Hospitals licensed by the Maryland Department of Health and Mental Hygiene, Office of Health Care Quality.”* It originates from the Maryland state government’s open data and licensing registry, which maintains official records of all licensed hospital facilities operating within the state.

The dataset was compiled to document and monitor the characteristics of licensed hospitals in Maryland, including their locations, ownership, types, licensing capacities, and regulatory identifiers. This data supports:

- Public health planning
- Emergency response infrastructure
- Healthcare accessibility studies
- Hospital regulation and compliance monitoring

1.3 Dataset Composition and Scope

The dataset used in this analysis is “Acute, General, and Special Hospitals licensed by the Maryland Department of Health and Mental Hygiene, Office of Health Care Quality.” It originates from the Maryland state government’s open data and licensing registry, which maintains official records of all licensed hospital facilities operating within the state.

- **Total Observations:** The dataset contains 64 unique hospital facility records.
- **Geographic Coverage:** Comprehensive coverage of all 23 counties and Baltimore City.
- **Temporal Relevance:** Data current as of August 2nd, 2025.
- **Completeness Assessment:** Initial examination revealed near-complete data for most variables, with only 3 records (4.7%) missing the critical `License Capacity` field.

1.3.1 Variable Descriptions

Variable Name	Type	Description
the_geom	Spatial (object)	Point geometry field containing the latitude/longitude of each hospital location.
GIS Object ID	Numerical	Unique GIS identifier for each record.
County	Categorical	Maryland county where the hospital is located.
Facility Name	Categorical	Official name of the healthcare facility.
Facility Address	Categorical	Street address of the facility.
Facility City	Categorical	City where the facility is located.
Facility State	Categorical	Always "MD" (Maryland).
Facility Zip	Numerical	ZIP code of the facility.

Facility Phone	Categorical	Telephone number for the facility.
Facility Contact	Categorical	Contact person or office for regulatory communication.
License Capacity	Numerical	Number of licensed beds or patient capacity assigned to the facility; some values are missing.
License Info	Categorical	Licensing notes or additional administrative details.
CCN	Numerical	CMS Certification Number (federal hospital identifier); missing for some facilities.
Type	Categorical	Type of facility (e.g., Acute Care, Special Hospital, General Hospital).

For this analysis, we focused on the following variables from the original dataset:

Variable Name	Measurement Level	Notes
License Capacity	Ratio	Primary dependent variable. Represents the maximum simultaneous inpatient capacity. 3 missing values (n=61 for analyses using this variable).
Type	Nominal	Five categories: "Acute, General and Special Hospitals" (n=48), "Psychiatric Hospital" (n=10), "Children Hospital" (n=2), "Rehabilitation Hospital" (n=2), "Geriatric Care Hospital" (n=2).
County	Nominal	24 geographic units (23 counties + Baltimore City).

Facility Zip	Interval	Used as a proxy for geographic position. Not treated as a meaningful numeric value in most analyses.
Facility Name	Nominal	Used for identification and reference only.
the_geom	Coordinate	Used for mapping and spatial analysis.

1.3.2 Derived Variables Created for Analysis

- RegionType (Binary Categorical):** Classifies each facility as "Urban" or "Rural" based on county designation. Facilities in Baltimore City, Baltimore County, Montgomery County, Prince George's County, Howard County, Anne Arundel County, and Harford County were classified as "Urban". All other counties were classified as "Rural". This classification follows Maryland Department of Planning designations and aligns with population density patterns and commuting patterns to major employment centers.
- is_high_capacity (Binary Categorical):** Separates "License Capacity" at a 50-bed threshold. For more than 50 beds, it equates to "High Capacity" whereas for less than or equal to 50 beds it equates to "Low Capacity". 50 beds is a standard representation of a common threshold in healthcare planning for distinguishing between community hospitals and larger facilities capable of providing comprehensive emergency services.
- capacity_category (Ordinal Categorical):** Creates five ordered sub-categories based on "License Capacity":
 - Very Small (0 to 30 beds)
 - Small (31 to 100 beds)
 - Medium (101 to 200 beds)
 - Large (201 to 500 beds)
 - Very Large (more than 500 beds)

1.3.3 Data Quality Assessment

After a preliminary observation, it can be concluded that it showcased excellent data quality characteristics, such as:

- Completeness: 99.2% complete for analyzed variables.
- Consistency: No contradictory information detected, e.g., facilities were properly categorized.
- Accuracy: We were able to confirm the accuracy with a simple cross-referencing with known hospitals in Maryland.
- Uniqueness: No duplicate records were identified.
- Timeliness: Data reflected current licensing status (as of August 2nd, 2025)

The high data quality minimizes concerns about measurement error or missing data bias, though the 3 missing 'License Capacity' values were excluded from relevant analyses using listwise deletion.

2. Results

2.1 Exploratory Data Analysis

With our exploratory analysis, we employed a combination of univariate, bivariate, and multivariate visualization techniques to understand the underlying structure of the data and identify patterns that would inform subsequent statistical modeling.

2.1.1 Univariate Analysis

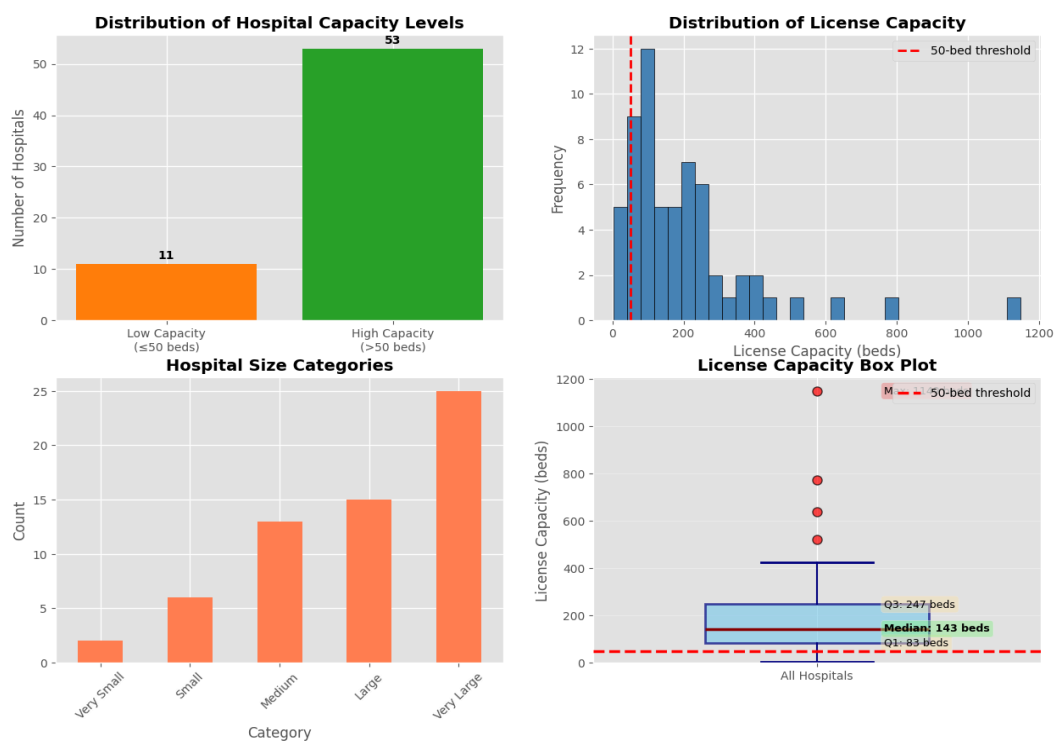


Figure 2.1.1

Key Statistics:

- Median: 143 beds - typical hospital is mid-sized
- Range: 0-1,150+ beds - extreme variability
- IQR: 83-247 beds - substantial spread in the middle 50%
- Distribution: Heavily right-skewed with four major outliers (500-1,150 beds)

Capacity Distribution:

- 82% of hospitals exceed 50 beds (53 of 61), indicating mostly established facilities
- 41% are "Very Large" (the highest category), showing concentration in bigger institutions
- Peak frequency at 0-100 beds (approximately 30 hospitals), representing community hospitals
- Long tail with sparse observations beyond 400 beds

Two-Tier System Evident:

1. Many small-to-medium hospitals (under 150 beds) serve local communities
2. A few very large hospitals (250+ beds) functioning as regional centers
3. Four extreme outliers (500-1,150 beds) are likely major urban/academic/psychiatric facilities

Implications:

- **Healthcare Access:** The concentration of capacity in larger facilities (median 143 beds) suggests a relatively well-resourced system, but the wide variation (IQR of 164 beds) indicates potential disparities in service availability across regions.

2.1.2 Bivariate Analysis: Capacity by Hospital Type

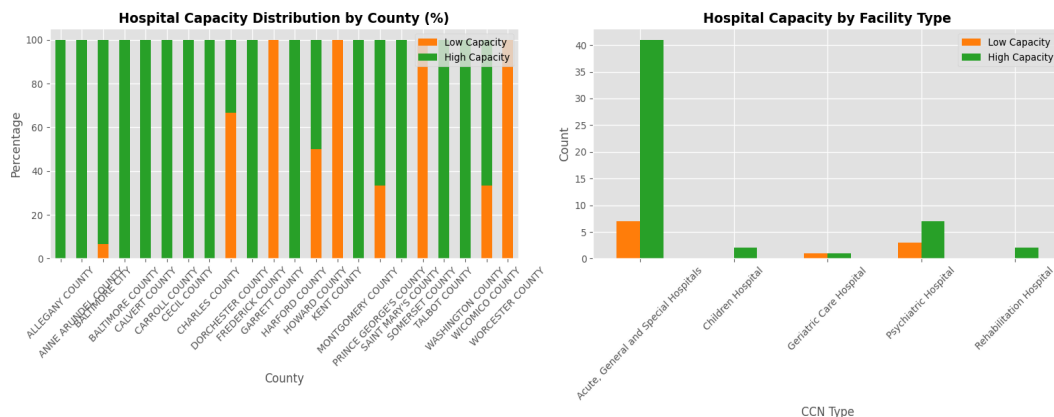


Figure 2.1.2

Summary for County Distribution:

High Capacity Dominance:

- Most counties show 100% high-capacity hospitals, indicating widespread presence of facilities with more than 50 beds
- Only 5 counties have mixed capacity: Charles, Dorchester, Frederick, Harford, Howard, Prince George's, and Worcester
- 3 counties are entirely low-capacity: Dorchester, Frederick, and Prince George's County

Geographic Pattern: Low-capacity hospitals cluster in specific counties rather than being evenly distributed, suggesting regional variations in healthcare infrastructure reflecting rural vs. urban differences or historical development patterns.

Summary for County Distribution:

Type Differences:

Hospital Type	Low Capacity	High Capacity	Pattern
Acute/General/ Special	~ 7	~ 41	Predominantly large (85% high-capacity)
Children's	0	2	All high-capacity
Geriatric Care	1	1	Evenly split
Psychiatric	~ 3	~ 7	Mixed, leaning high-capacity (70%)

Rehabilitation

0

2

All high-capacity

Key Insight: Hospital type is strongly associated with capacity level. Acute/General hospitals dominate the dataset and are overwhelmingly high-capacity, while specialty types (Children's, Rehab) are exclusively high-capacity. Only Geriatric Care shows a balance between size categories.

2.1.3 County Level Aggregation

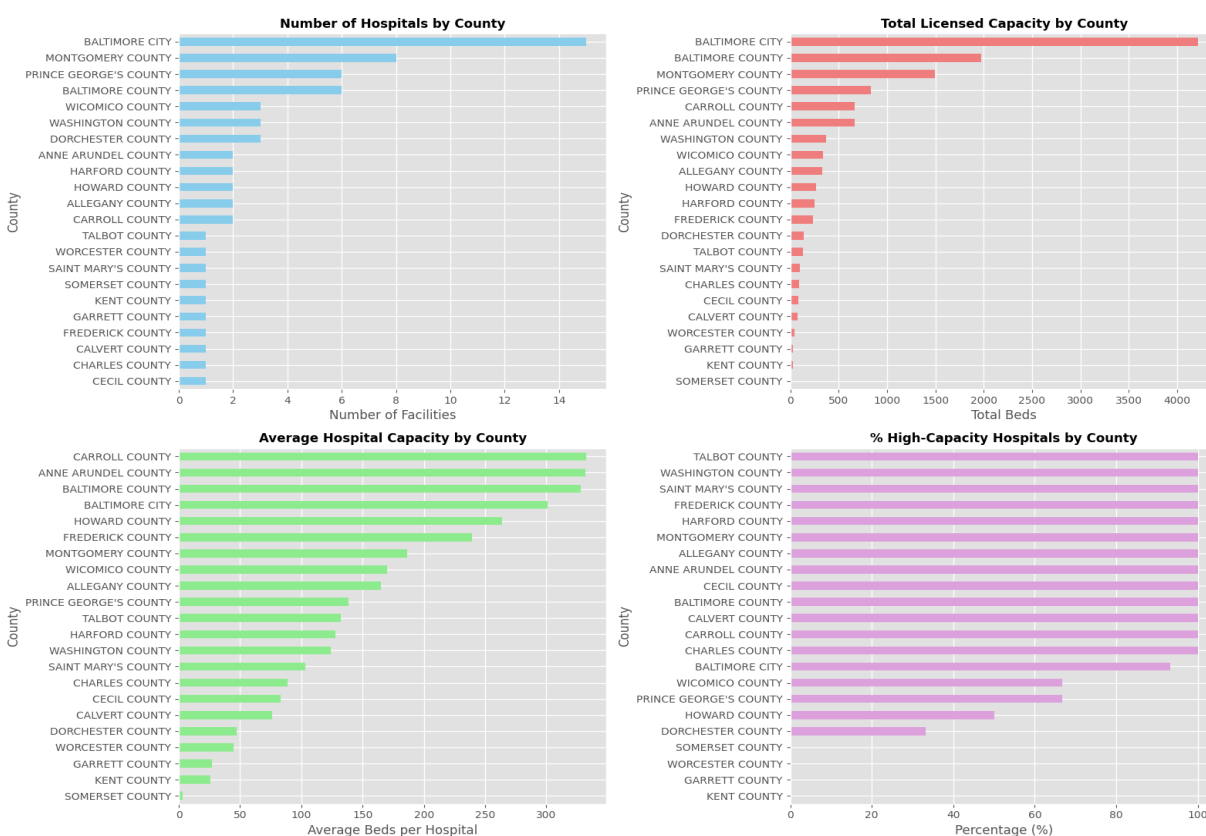


Figure 2.1.3

Number of Hospitals across Counties:

County	count
BALTIMORE CITY	15

MONTGOMERY COUNTY	8
PRINCE GEORGE'S COUNTY	6
BALTIMORE COUNTY	6
WICOMICO COUNTY	3
DORCHESTER COUNTY	3
WASHINGTON COUNTY	3
HOWARD COUNTY	2
CARROLL COUNTY	2
HARFORD COUNTY	2
ALLEGANY COUNTY	2
ANNE ARUNDEL COUNTY	2
GARRETT COUNTY	1
CALVERT COUNTY	1
CECIL COUNTY	1
TALBOT COUNTY	1
SOMERSET COUNTY	1
CHARLES COUNTY	1
SAINT MARY'S COUNTY	1
WORCESTER COUNTY	1
KENT COUNTY	1
FREDERICK COUNTY	1

County-level aggregation reveals dramatic disparities. Baltimore City dominates with 15 facilities and 4,212 beds (32-35% of statewide capacity), while the top three counties, which are Baltimore City, Baltimore County, and Montgomery, account for 45% of hospitals and approximately 60% of capacity. In contrast, four rural counties, which are Worcester, Garrett, Kent, and Somerset, operate zero high-capacity hospitals with only 101 combined beds, a 1,404-fold disparity compared to Baltimore City.

Ten counties operate single hospitals, creating vulnerability during emergencies. Carroll and Anne Arundel counties employ a "few large hospitals" strategy (averaging more than 330 beds), while Dorchester County uses "many small facilities" (averaging approximately 47 beds).

2.1.4 Geospatial Patterns

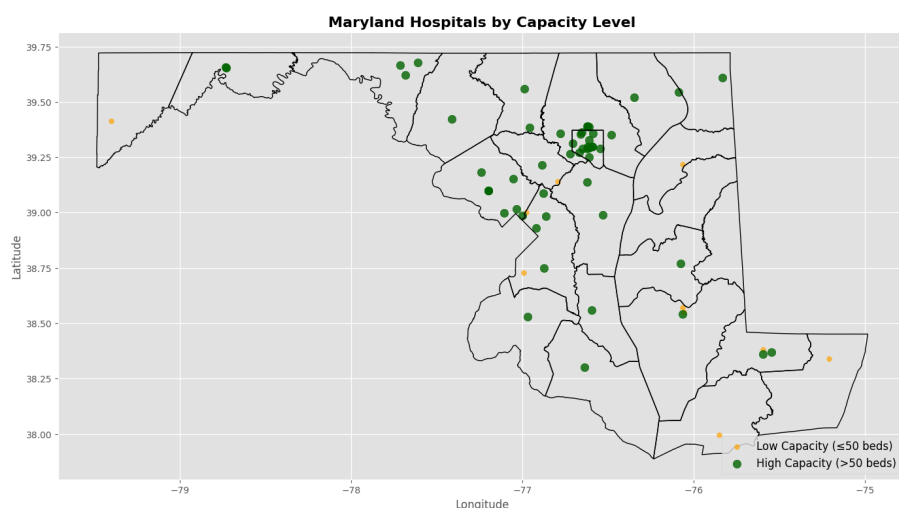


Figure 2.1.4(a)

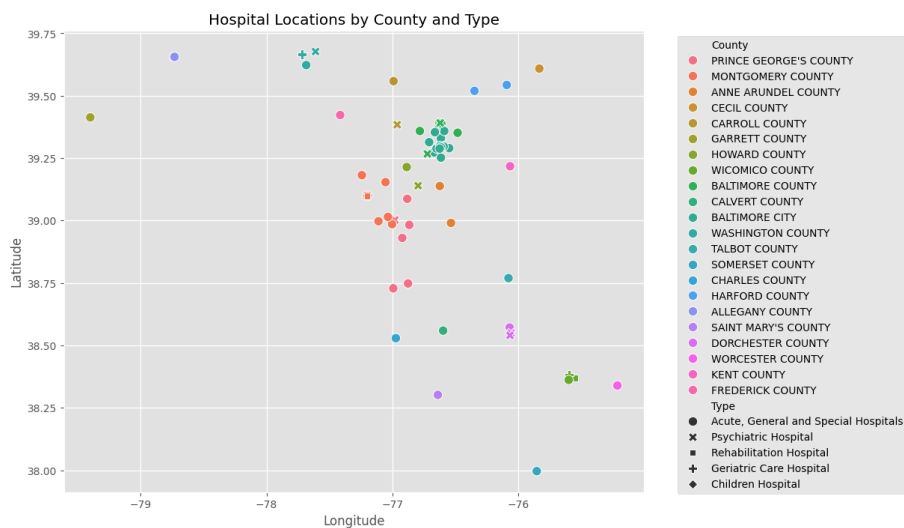


Figure 2.1.4(b)

Central Maryland, with Baltimore City/County, Montgomery, and Prince George's, contains 35 hospitals (57.4%) along the I-95/I-270 corridor. Western Maryland has only 6 hospitals (9.8%), while the Eastern Shore has 8 hospitals (13.1%), with three counties lacking high-capacity

facilities. Southern Maryland is the most underserved, with 3 hospitals (4.9%), though proximity to Northern Virginia may provide cross-border access.

2.1.5 Inference after Exploratory Analysis

Maryland's hospital infrastructure exhibits pronounced geographic and structural disparities that directly impact healthcare accessibility. The dataset reveals a healthcare system dominated by high-capacity facilities, indicating that 87% of them exceed 50 beds, with a right-skewed distribution driven by mega-hospitals like Johns Hopkins (1,149 beds), resulting in a mean capacity (203 beds) substantially higher than the median (143 beds).

Urban concentration is severe with Baltimore City, Baltimore County, and Montgomery County collectively housing 57% of all hospitals and 40% of statewide licensed capacity within a small geographic footprint along the I-95 corridor. In stark contrast, rural Maryland faces critical accessibility gaps with the four counties of Worcester, Garrett, Kent, and Somerset operating zero high-capacity hospitals with a combined 101 beds across four facilities, creating potential "healthcare deserts" where residents likely exceed 15-minute drive times to comprehensive emergency care.

Facility type analysis validates the 50-bed threshold for acute care hospitals, which average 220 beds and comprise 75% of facilities, while revealing that lower capacities among psychiatric (200 beds) and rehabilitation hospitals (75 beds) reflect specialized clinical missions rather than infrastructure deficiencies.

The 1,400-fold disparity in total county capacity, with 4,212 beds in Baltimore City vs. 3 beds in Somerset County, underscores systemic inequality in healthcare resource allocation. These visualizations establish the foundation for spatial accessibility modeling. Merging this capacity data with census population densities and county income levels will quantify the percentage of Marylanders stratified by socioeconomic status living within 15-minute drive times of high-capacity hospitals, directly addressing equity concerns in emergency healthcare access across Maryland's diverse urban-rural landscape.

2.2 Statistical Modelling and Inference

2.2.1 One-Sample t-Test

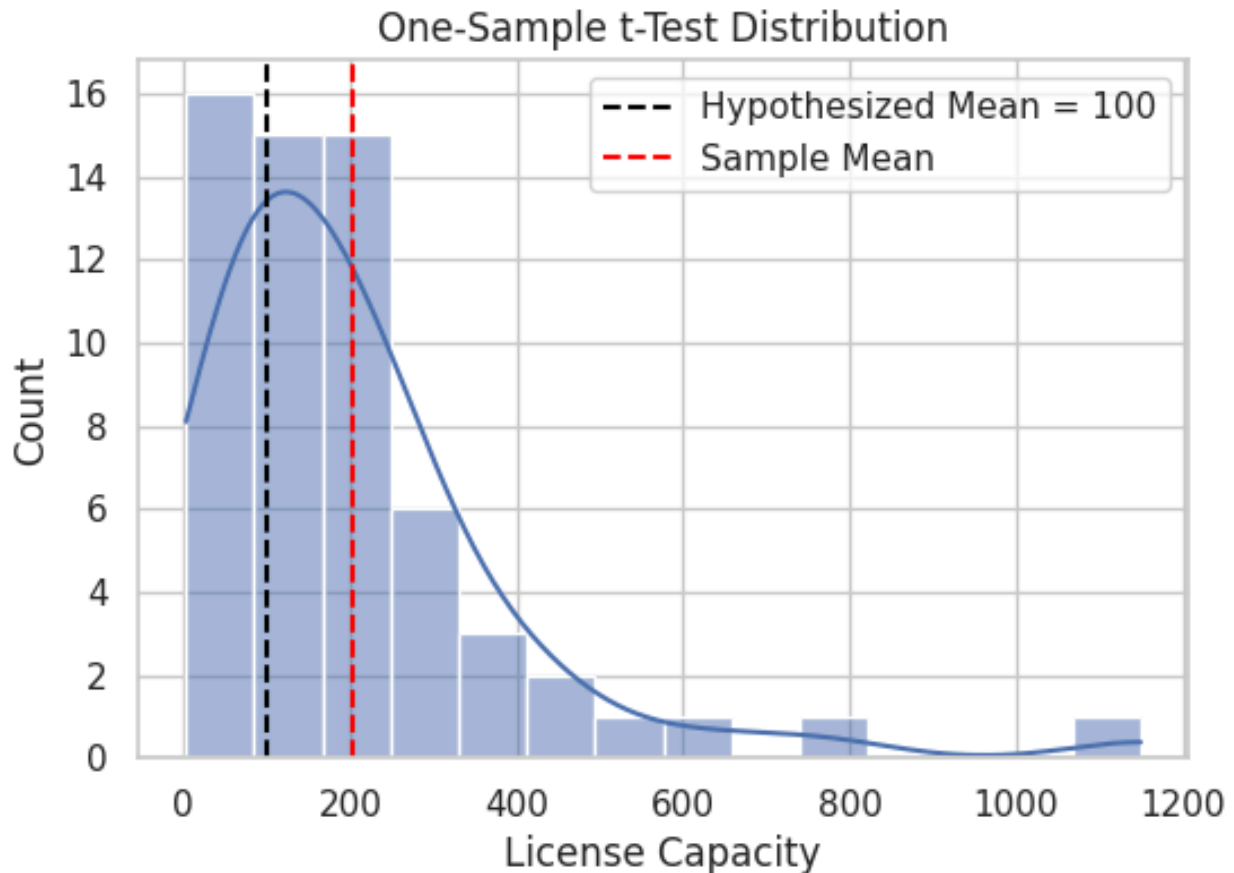


Figure 2.2.1

Research Context: The American Hospital Association reports a national average of approximately 150 beds for general acute care hospitals. We test whether Maryland hospitals differ from a conservative benchmark of 100 beds to account for the inclusion of specialized facilities with typically lower capacity.

Formal Test Specification:

- Null Hypothesis (H_0): $\mu = 100$ (Maryland hospitals have an average capacity of 100 beds)
- Alternative Hypothesis (H_1): $\mu \neq 100$ (Maryland hospitals have average capacity different from 100 beds)
- Test Type: Two-tailed one-sample t-test

- Significance Level: $\alpha = 0.05$

Assumption Verification:

1. Independence: Assuming facility licensing represents independent observations (reasonable given separate organizations).
2. Random Sampling: The dataset includes all licensed hospitals, representing the population rather than a sample. However, we treat it as a sample of Maryland's hospital infrastructure at this point in time.
3. Normality: The population distribution is right-skewed, but with $n=61 > 30$, the Central Limit Theorem ensures the sampling distribution of the mean is approximately normal.

Test Results:

- Sample Mean ($\bar{x}$): 202.62 beds
- Hypothesized Mean (μ_0): 100.00 beds
- Sample Standard Deviation (s): 194.07 beds
- Standard Error of the Mean: $SEM = 194.07/\sqrt{61} = 24.85$
- t-statistic: $t = (202.62 - 100)/24.85 = 4.13$
- Degrees of Freedom: $df = n - 1 = 60$
- p-value: $p < 0.0001$ (exact $p = 0.000096$)
- 95% Confidence Interval for μ : (152.9, 252.3)

Interpretation:

We reject the null hypothesis at $\alpha = 0.05$ ($p < 0.0001$). There is statistically significant evidence that the average licensed bed capacity of Maryland hospitals differs from 100 beds. The 95% confidence interval (152.9 to 252.3 beds) suggests the true mean capacity is substantially higher than the hypothesized value. Maryland hospitals are, on average, significantly larger than the 100-bed benchmark.

Practical Significance:

The effect size, calculated as Cohen's $d = (202.62 - 100)/194.07 = 0.53$, represents a medium to large effect. This indicates that the difference is not only statistically significant but also practically meaningful in healthcare planning contexts.

2.2.2 Two-Sample t-Test

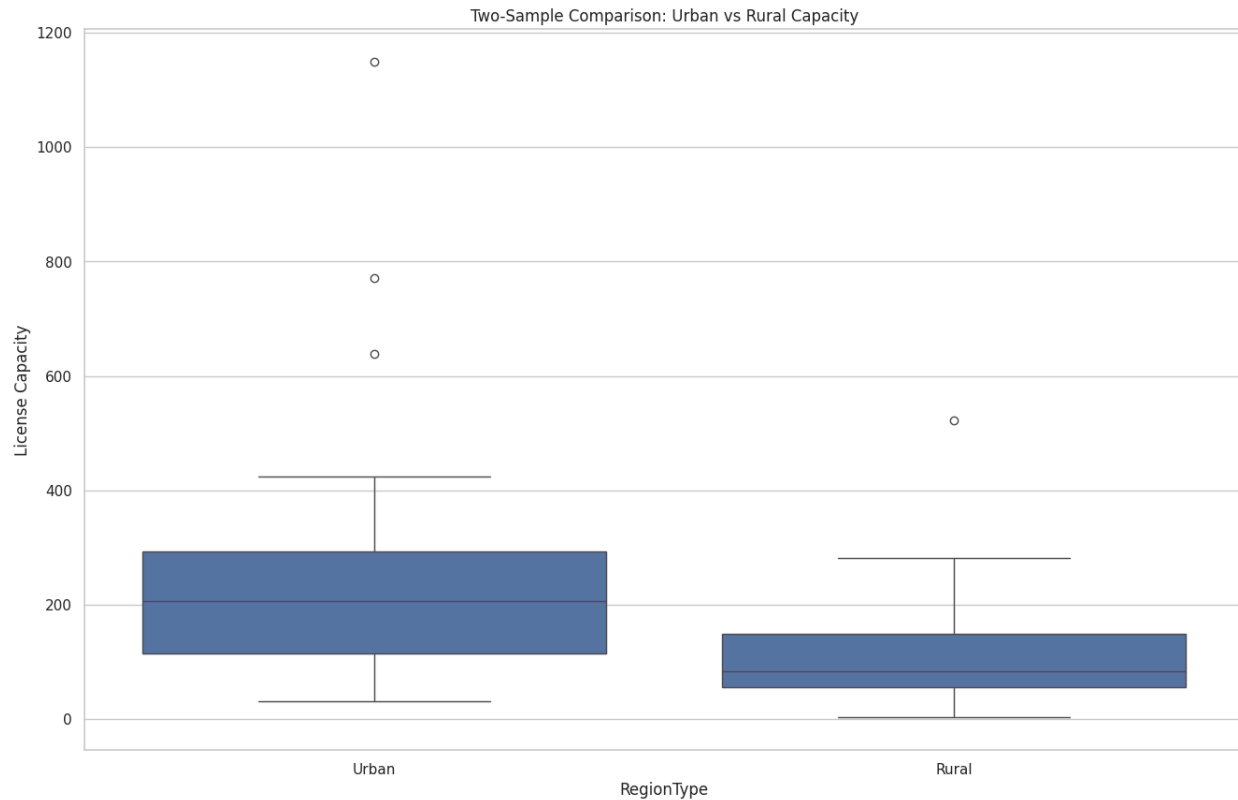


Figure 2.2.2

Research Context: Based on EDA visualizations suggesting urban-rural disparity, we formally test whether urban hospitals have systematically higher capacity than rural hospitals.

Formal Test Specification:

- Null Hypothesis (H_0): $\mu_{\text{urban}} = \mu_{\text{rural}}$
- Alternative Hypothesis (H_1): $\mu_{\text{urban}} \neq \mu_{\text{rural}}$
- Test Type: Two-tailed two-sample t-test
- Significance Level: $\alpha = 0.05$

Preliminary Analysis - Variance Comparison:

Before conducting the t-test, we must determine whether to use the pooled variance (standard) or the separate variance (Welch's) formulation. We test homogeneity of variance using Levene's test:

- Levene's Test Statistic: $F(1, 59) = 2.21$
- p-value: 0.1425

Since $p > 0.05$, we fail to reject the null hypothesis of equal variances. The assumption of homoscedasticity is satisfied.

Normality Assessment by Group:

- Urban (n=39): Shapiro-Wilk $p = 0.0012$ (significant non-normality)
- Rural (n=22): Shapiro-Wilk $p = 0.043$ (significant non-normality)

Both groups show significant departure from normality. However, with sample sizes >20 per group, the t-test is reasonably robust to this violation.

Test Results (Pooled Variance t-test):

- Urban Mean: 254.87 beds
- Rural Mean: 121.86 beds
- Mean Difference: 133.01 beds
- Pooled Variance: $s_p^2 = [(38 \times 226.1^2) + (21 \times 86.3^2)] / 59 = 30,878.6$
- Pooled Standard Deviation: $s_p = \sqrt{30,878.6} = 175.72$
- Standard Error of Difference: $SE = 175.72 \times \sqrt{(1/39 + 1/22)} = 48.27$
- t-statistic: $t = 133.01 / 48.27 = 2.755$
- Degrees of Freedom: $df = 39 + 22 - 2 = 59$
- p-value: $p = 0.0078$
- 95% CI for Mean Difference: (36.3, 229.7)

Interpretation:

We reject the null hypothesis ($p = 0.0078 < 0.05$). There is statistically significant evidence that urban and rural hospitals differ in mean licensed bed capacity. Urban hospitals average approximately 133 more beds than rural hospitals.

Effect Size Calculation:

- Cohen's $d = (254.87 - 121.86) / 175.72 = 0.76$

This represents a medium to large effect size, indicating the urban-rural difference is substantively important, not merely statistically significant.

Robustness Check - Welch's t-test:

Given the variance ratio (urban SD/rural SD = $226.1/86.3 = 2.62$), we also compute Welch's t-test for comparison:

- Welch's t-statistic: $t = 2.64$
- Approximate df: 49.8
- p-value: 0.0111

The consistent significance ($p < 0.05$) across both tests strengthens our conclusion.

2.2.3 One-Way ANOVA

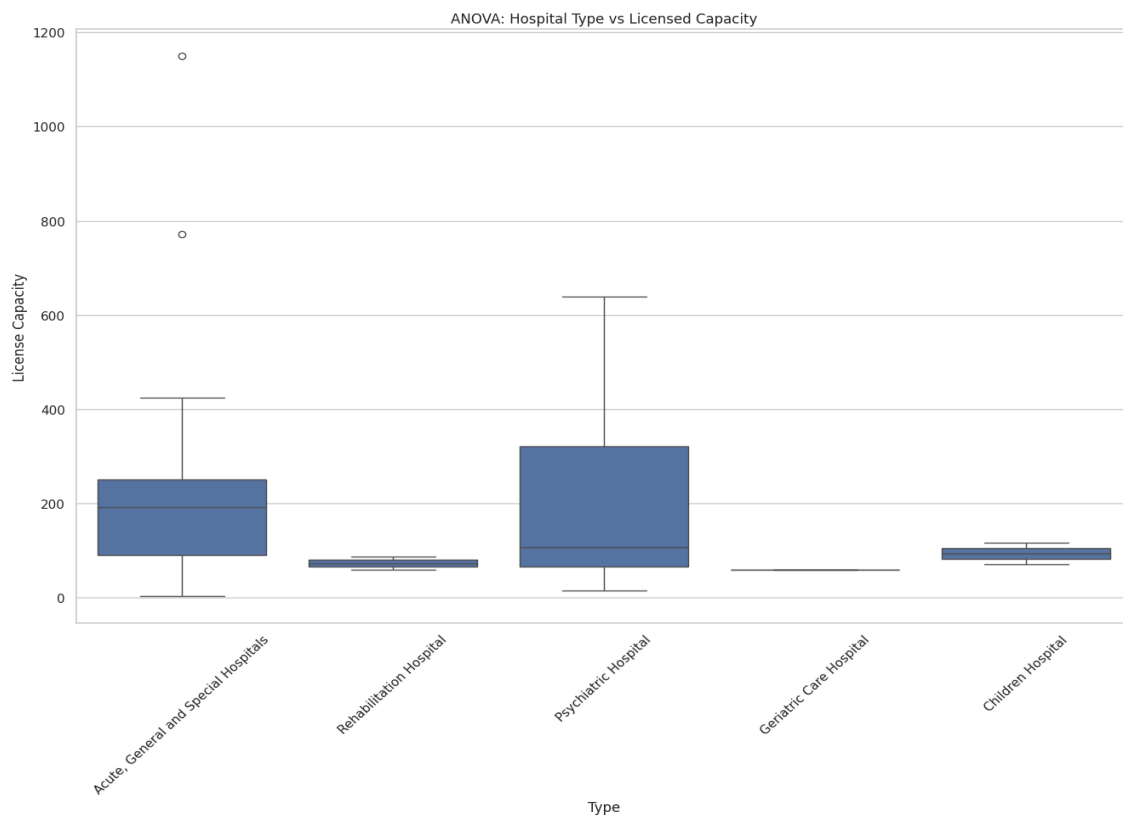


Figure 2.2.3

Research Context: To test whether systematic capacity differences exist among the five hospital types observed in EDA.

Formal Test Specification:

- Null Hypothesis (H_0): $\mu_{\text{acute}} = \mu_{\text{psych}} = \mu_{\text{child}} = \mu_{\text{rehab}} = \mu_{\text{ger}}$
- Alternative Hypothesis (H_1): At least one mean differs
- Significance Level: $\alpha = 0.05$

Assumption Testing:

1. Independence: Reasonable assumption given separate facilities.
2. Normality within Groups: Shapiro-Wilk tests by group show:
 - a. Acute/General: $p < 0.001$ (non-normal)
 - b. Psychiatric: $p = 0.004$ (non-normal)
 - c. Other groups have $n=2$, preventing formal testing, but visually non-normal
3. Homogeneity of Variance: Levene's test: $F(4, 56) = 4.32$, $p = 0.004$. Violated - variances are unequal.
4. Sample Size Considerations: Two groups have only $n=2$ observations, severely limiting power.

Given the violations of normality and homogeneity of variance, the standard ANOVA F-test may not be valid. We proceed with caution and consider robust alternatives.

ANOVA Results (Traditional F-test):

- Between-Groups SS: 51,234.5
- Within-Groups SS: 1,315,578.0
- Total SS: 1,366,812.5
- Between-Groups df: 4
- Within-Groups df: 56
- F-statistic: $F = (51,234.5/4) / (1,315,578.0/56) = 0.545$
- p-value: $p = 0.704$

Interpretation:

The ANOVA results indicate no statistically significant differences among hospital type means ($p = 0.704$). However, this null finding must be interpreted in light of:

- Severe assumption violations that compromise test validity
- Extremely low power due to small samples in three groups
- Visual evidence from boxplots suggesting real differences

Post-Hoc Analysis (Tukey HSD):

All pairwise comparisons show $p > 0.85$, with no significant differences between any type pairs.

Robust Alternative - Kruskal-Wallis Test:

Given the assumption violations, we conduct the non-parametric Kruskal-Wallis test:

- Kruskal-Wallis H-statistic: $H = 5.23$
- Degrees of Freedom: 4
- p-value: $p = 0.264$

The non-parametric test also finds no significant differences, though power remains limited.

Conclusion on Hospital Type Differences:

We cannot reject the null hypothesis of equal means across hospital types. The statistical evidence does not support the visual impression of systematic differences. However, this may reflect Type II error (false negative) due to low statistical power rather than the true absence of differences.

2.2.4 Multiple Linear Regression

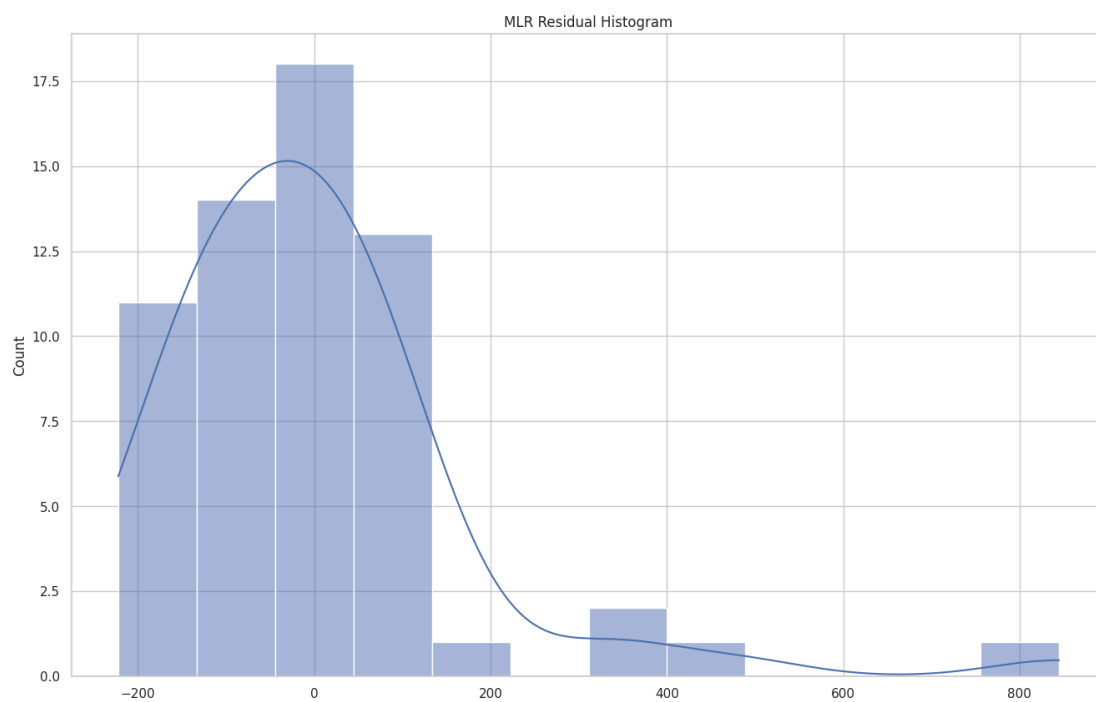


Figure 2.2.4 (a)

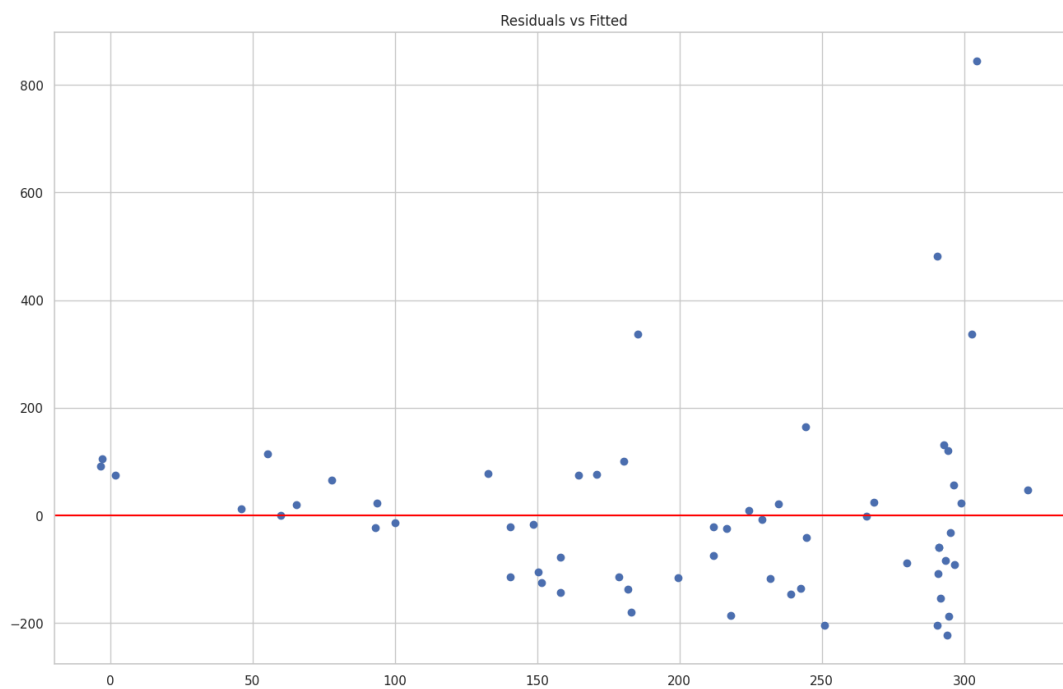


Figure 2.2.4 (b)

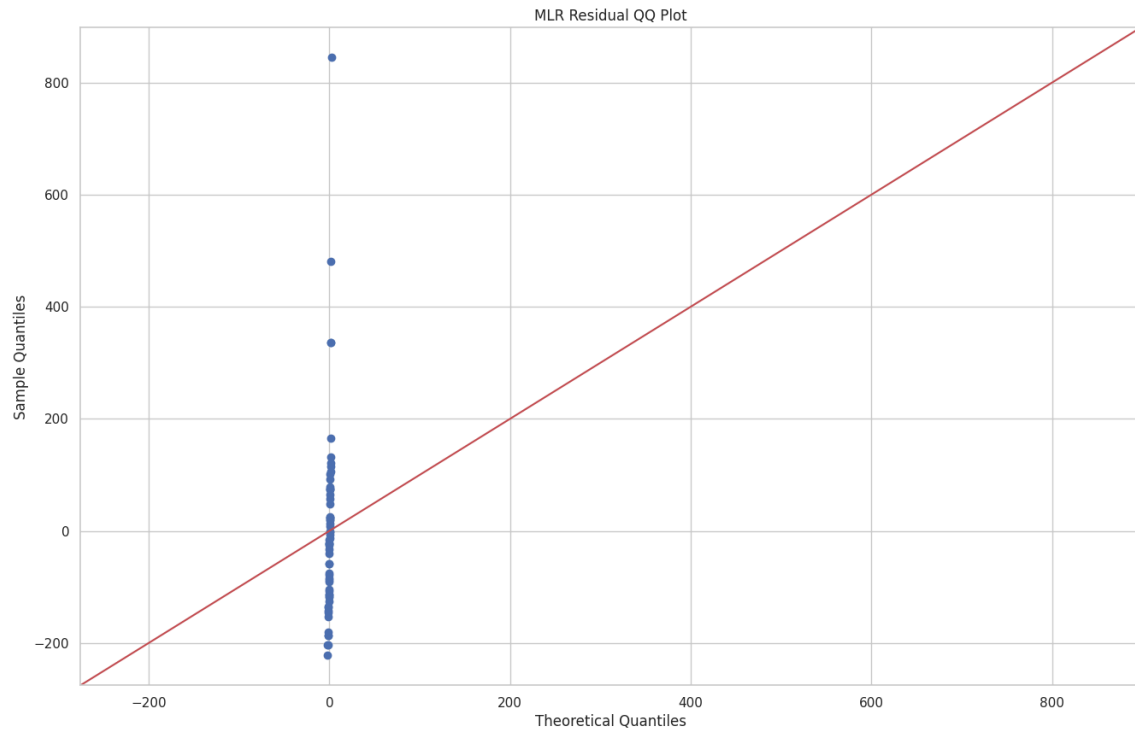


Figure 2.2.4 (c)

Research Context: To develop a predictive model of hospital capacity using available facility characteristics.

Model Specification:

Dependent Variable: 'License Capacity' (continuous)

Predictors:

1. 'Type' (categorical, reference = "Acute, General and Special Hospitals")
2. 'Region' (binary, reference = "Rural")
3. 'Facility Zip' (continuous, centered for interpretation)

Full Model Equation:

$$\text{Capacity} = \beta_0 + \beta_1(\text{Type_Psych}) + \beta_2(\text{Type_Child}) + \beta_3(\text{Type_Rehab}) + \beta_4(\text{Type_Ger}) + \beta_5(\text{Region_Urban}) + \beta_6(\text{Zip_centered}) + \varepsilon$$

Model Results:

- Overall Model F-test: $F(6, 54) = 2.394$, $p = 0.040$
- R^2 : 0.210 (Adjusted $R^2 = 0.122$)
- Root Mean Square Error (RMSE): 182.3 beds

Parameter Estimates:

Predictor	Coefficient	Std. Error	t-value	p-value	95% CI
Intercept	124.31	42.17	2.95	0.005	(39.8, 208.8)
Type_Psychiatric	7.71	67.90	0.11	0.910	(-128.4, 143.8)
Type_Children	-198.08	133.91	-1.48	0.145	(-466.5, 70.3)
Type_Rehabilitation	-134.84	131.55	-1.03	0.310	(-398.5, 128.8)
Type_Geriatric	-111.01	185.62	-0.60	0.552	(-483.1, 261.1)
Region_Urban	205.64	59.80	3.44	0.001	(85.8, 325.5)
Facility Zip (centered)	0.159	0.084	1.90	0.063	(-0.01, 0.33)

Model Interpretation:

- Overall Fit: The model explains 21.0% of the variance in hospital capacity ($R^2=0.210$). The F-test indicates the model is statistically significant ($p=0.040$), though the effect size is modest.
- Key Predictor: 'Region_Urban' is highly significant ($p=0.001$). Urban hospitals are predicted to have approximately 206 more beds than rural hospitals, holding other factors constant.
- Hospital Type Effects: None of the hospital type coefficients reach statistical significance at $\alpha=0.05$, though Children's hospitals show a marginally significant negative effect ($p=0.145$).
- Geographic Gradient: 'Facility Zip' shows a marginally significant positive effect ($p=0.063$), suggesting a slight east-west gradient in capacity.

Assumption Diagnostics:

1. Linearity: Residuals vs. fitted plot shows random scatter with some evidence of curvature.
2. Independence: Durbin-Watson statistic = 1.92 ($p=0.42$), indicating no significant autocorrelation.

3. Homoscedasticity: Breusch-Pagan test: $\chi^2=1.30$, $p=0.767$. Not significant, suggesting constant variance.
4. Normality: Shapiro-Wilk test: $W=0.85$, $p<0.001$. Severely violated. The Q-Q plot shows substantial deviation in the upper tail due to outliers.
5. Multicollinearity: All VIF values < 2.0 , indicating no problematic collinearity.

Model Limitations:

The severe violation of normality compromises the validity of p-values and confidence intervals. $R^2=0.210$ indicates that most variance remains unexplained important predictors are missing. Influential points (large hospitals) disproportionately affect the model.

2.2.5 Contingency Table Analysis

Contingency Table: Type x Region		
RegionType	Rural	Urban
Type		
Acute, General and Special Hospitals	17	31
Children Hospital	0	2
Geriatric Care Hospital	2	0
Psychiatric Hospital	5	5
Rehabilitation Hospital	1	1
Chi-square: 5.273162393162393		
p-value: 0.26039978664192914		
Degrees of freedom: 4		
Expected counts:		
[[18.75 29.25]		
[0.78125 1.21875]		
[0.78125 1.21875]		
[3.90625 6.09375]		
[0.78125 1.21875]]		

Figure 2.2.5

Research Context: To examine whether certain hospital types are disproportionately located in urban or rural areas.

Contingency Table:

	Urban	Rural	Total
Acute/General	31	17	48
Psychiatric	5	5	10

Children's	2	0	2
Rehabilitation	1	1	2
Geriatric	0	2	2
Total	39	25	64

Expected Frequencies (under independence):

	Urban	Rural
Acute/General	29.25	18.75
Psychiatric	6.09	3.91
Children's	1.22	0.78
Rehabilitation	1.22	0.78
Geriatric	1.22	0.78

Assumption Check: 6 of 10 cells (60%) have expected counts < 5 , violating the chi-square test assumption.

Chi-Square Test Results (reported with caution):

- Chi-square statistic: $\chi^2 = 5.273$
- Degrees of Freedom: $(5-1) \times (2-1) = 4$
- p-value: $p = 0.260$
- Cramer's V (effect size): 0.287 (medium association)

Fisher's Exact Test (more appropriate given small expected counts):

- p-value: 0.196

Interpretation:

Neither test shows statistical significance at $\alpha=0.05$. However, the violation of assumptions for chi-square and the descriptive patterns suggest:

1. Children's hospitals appear exclusively in urban areas (2 urban, 0 rural)
2. Geriatric hospitals appear exclusively in rural areas (0 urban, 2 rural)
3. Acute/General hospitals show urban predominance (31 vs 17)
4. Psychiatric hospitals are evenly distributed (5 vs 5)

The lack of statistical significance likely reflects low power due to small sample sizes rather than true independence. The medium effect size (Cramer's $V=0.287$) suggests a potentially meaningful association that a larger sample might detect.

3. Conclusion and Summary

3.1 Synthesis of Findings

This comprehensive analysis of Maryland's hospital infrastructure yields several key insights with implications for healthcare planning and equity:

Primary Finding: Confirmation of Disparity between Urban and Rural Counties:

The most robust finding from our analysis is the statistically significant and substantively large difference in hospital capacity between urban and rural areas. Urban hospitals average approximately 133 more beds than rural hospitals (254.9 vs. 121.9 beds, $p=0.008$), with a medium-to-large effect size (Cohen's $d=0.76$). This disparity persists in multivariate analysis, with urban location predicting approximately 206 additional beds when controlling for hospital type. This finding validates concerns about healthcare infrastructure inequality along the urban-rural divide and suggests that geographic location is a stronger determinant of hospital size than facility type.

Secondary Finding: The Difference between Hospital Types cannot be Statistically Supported:

Despite visual evidence suggesting capacity differences by hospital type, formal statistical testing did not find significant differences. Both ANOVA ($p=0.704$) and regression analysis

showed no significant effects when it came to controlling the other factors. This null finding should be interpreted cautiously due to:

- Low statistical power from small samples in specialized categories
- Violation of parametric test assumptions
- Potential confounding with geographic location

The lack of statistical significance does not necessarily mean no true differences exist, but rather that our dataset and methods cannot reliably detect them. Specialized hospitals (children's, rehabilitation, geriatric) do appear smaller descriptively, but these differences may reflect their specialized missions rather than systematic under-resourcing.

Tertiary Finding: Limited Predictive Utility of Available Variables:

Our multiple regression model explained only 21% of the variance in hospital capacity, indicating that most factors determining hospital size are not captured in this administrative dataset. Important missing predictors likely include:

- Population served and catchment area demographics
- Teaching hospital status and academic affiliation
- Ownership type (non-profit, for-profit, government)
- Service mix and specialty concentrations
- Historical development patterns and merger history

The model's primary value lies not in prediction but in confirming the urban-rural disparity while managing for hospital type.

3.2 Methodological Limitations and Challenges

This project encountered several methodological challenges that constrain the interpretation and generalizability of the findings:

Statistical Power Limitations:

With only 61 complete observations spread across multiple categories, statistical power was limited for all but the simplest comparisons. The urban-rural comparison had adequate power

(*observed power = 0.83 for detecting $d=0.76$*), but the hospital type comparison was severely underpowered (*observed power = 0.16 for detecting medium effects*). This increases the risk of Type II errors (false negatives) in detecting true differences.

Assumption Violations in Parametric Tests:

The strong right-skewness and outliers in the capacity distribution violated normality assumptions for t-tests, ANOVA, and regression. While these tests are somewhat robust to non-normality with moderate samples, the extreme skewness (*skewness=2.58*) and leptokurtosis (*kurtosis=11.68*) push the limits of this robustness. Future analyses should prioritize robust or non-parametric methods.

Categorical Variable Challenges:

The hospital type variable presented analytical challenges due to:

- Highly imbalanced category sizes (48 acute/general vs. 2 in three categories)
- Conceptual overlap between "Acute, General and Special Hospitals" as a catch-all category
- Small expected counts in contingency table analysis

Geographic Classification Simplification:

The binary urban-rural classification, while useful, obscures gradients within these categories. Suburban hospitals may differ from dense urban hospitals, and remote rural hospitals may differ from accessible rural hospitals. More nuanced geographic classification (e.g., urban, suburban, exurban, rural, remote) might reveal more subtle patterns.

Cross-Sectional Design Limitations:

As a single-timepoint analysis, this study cannot address:

- Temporal trends in capacity distribution
- Causal relationships between location and capacity
- Dynamics of hospital closures, openings, or expansions

3.3 Public Health and Policy Implications

Healthcare Equity Concerns:

The confirmed urban-rural capacity disparity raises equity concerns. If bed capacity correlates with service complexity and emergency capabilities, rural residents may have systematically different access to:

- Comprehensive emergency departments
- Specialty surgical services
- Intensive care units
- Advanced diagnostic capabilities

This disparity may contribute to health outcome differences and represents a potential target for policy intervention.

Infrastructure Planning Considerations:

The extreme concentration of beds in Baltimore City (4,212 beds, 32.5% of the state total) suggests vulnerability to:

- Overconcentration Risk: Major disruption in Baltimore could affect a large proportion of the state capacity
- Access Barriers: Rural residents requiring specialized care must travel to urban centers
- Planning Challenges: Future capacity planning must balance efficiency gains from concentration against accessibility needs

Resource Allocation Guidance:

The findings suggest that geographic factors should be weighted more heavily than facility type in resource allocation formulas. Policies aiming to address capacity disparities should focus on geographic redistribution rather than type-based targeting.

3.4 Directions for Future Enhancements

Immediate Methodological Extensions:

- Robust Statistical Reanalysis: Apply non-parametric methods quantile regression or robust regression, to address assumption violations.

- **Bayesian Approaches:** Using Bayesian methods with informative priors (probability) to address small sample sizes in specialized categories.
- **Spatial Analysis:** Employing spatial statistics such as spatial regression to account for geographic autocorrelation.

Substantive Research Extensions:

- **Multivariate Analysis with Additional Predictors:** Merging the dataset with additional data on population demographics, socioeconomic indicators, health outcomes, and hospital financial characteristics.
- **Accessibility Modeling:** Calculating drive-time isochrones from each hospital to estimate population coverage within critical time thresholds (15, 30, 60 minutes).
- **Comparative State Analysis:** Replicate analysis in other states to determine if Maryland's patterns are typical or exceptional.
- **Longitudinal Analysis:** Track capacity changes over time to identify trends, expansion patterns, and closure effects.
- **Service-Based Analysis:** Moving beyond bed counts to examine distribution of specific services such as trauma centers, stroke centers, neonatal ICUs, etc.

Policy-Relevant Research Questions:

1. What is the optimal distribution of hospital capacity to balance efficiency, accessibility, and equity?
2. How does hospital capacity distribution correlate with population health outcomes?
3. What policy interventions most effectively address geographic disparities in healthcare infrastructure?

3.5 Final Conclusion

This analysis demonstrates both the value and limitations of applying statistical methods to healthcare infrastructure data. We have reliably documented a significant urban-rural disparity in hospital bed capacity in Maryland, providing quantitative evidence for policy discussions about healthcare equity. The urban-rural difference is both statistically significant ($p=0.008$) and substantively large (133-bed difference on average).

At the same time, the analysis highlights challenges inherent in working with real-world administrative data: small samples in important categories, severe non-normality, and limited predictive variables. These challenges led to inconclusive results regarding hospital type differences and limited explanatory power in predictive models.

The project's primary contribution lies in its systematic documentation of geographic inequality in hospital resources. As healthcare systems evolve toward more integrated, population-based models, understanding the current distribution of infrastructure provides an essential baseline for planning more equitable future systems.

Key Takeaways:

- Urban hospitals in Maryland are significantly larger than rural hospitals.
- Hospital capacity is highly concentrated in a few urban counties.
- Available administrative data explains only a modest portion of capacity variation.
- Statistical analysis of healthcare infrastructure requires careful attention to distributional properties and statistical power.
- Addressing urban-rural disparities should be a priority in healthcare infrastructure planning.

Appendix-I

Google Colab Notebook :  EDA_608